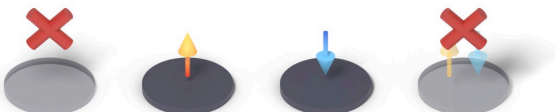
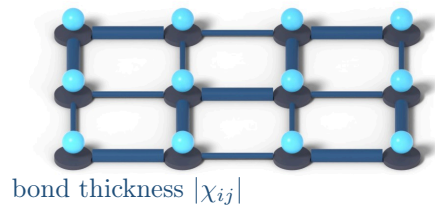
(a)


$|0\rangle$
$f_\uparrow^\dagger|0\rangle$
$f_\downarrow^\dagger|0\rangle$
$f_\uparrow^\dagger f_\downarrow^\dagger|0\rangle$

(b)


bond thickness $|\chi_{ij}|$